A bike nearby. A bike available?

Is the supply of public bikes proportional to effective demand? We test it, then ask what a fixed fleet could do about the gap.

> **H_0** Supply is proportional to effective demand across areas.
> **H_1** At least one area's supply share differs from its effective demand share.

ANSWER

We reject H_0 at the station level, by a wide margin. An HC3 Wald test of the zero-intercept restriction rejects strict proportionality in both weekparts. Nominal docks rise with predicted demand, but at under a fifth of the proportional rate, and about 10 docks exist at zero demand. **At district level the gap is milder:** across 25 districts about 10% of docks would have to move to match demand shares, and no district is more than 2.7 points off. The result is exploratory (predicted demand is a proxy) and describes association, not cause.

4.6 vs 23.2
docks per +100 requests/day: fitted vs proportional (weekday)

9.8
docks at zero demand (proportional: 0)

R^2 0.13
87% of dock variation unexplained (weekend R^2 0.09)

18.3%
of weekday bikes would move in the fixed-fleet plan

METHOD

- **Data.** 2,695 stations, 80,601 station-day rows, one month. Predicted daily demand, nominal docks, midnight stock and stockout rates. 223 weekday and 224 weekend stations were left out: inactive, depot or repair, or dock and inventory counts that disagree.
- **Tests.** OLS of docks on predicted demand per 100 requests with HC3 robust errors, against the proportional line through zero with slope k = total docks ÷ total demand. Dissimilarity index and chi-square describe the gap; they are not significance claims.
- **Plan.** Need = demand ÷ bike turnover. A water-filling step shares the existing fleet by need, limited by a dock ceiling, so no bikes are added. Five sensitivity variants and data flags give each station a HIGH, MEDIUM or LOW rating.

TEST RESULTS

TEST	WEEKDAY	WEEKEND
Slope, docks per +100 requests/day (95% CI)	4.58 (3.63 to 5.53)	4.27 (3.48 to 5.05)
Proportional slope k	23.2	27.8
Intercept, docks at zero demand (95% CI)	9.79 (9.32 to 10.25)	10.32 (9.98 to 10.67)
HC3 Wald $\chi^2(1)$, intercept = 0 (nonredundant H_0 test)	1,731.7, p < 0.001	3,463.3, p < 0.001
Joint $\chi^2(2)$, intercept = 0 and slope = k (k fixed; exploratory)	1,732.7	3,810.6
Slope = 0	p < 1e-15	p < 1e-15
Proportional line vs fitted line, residual sum of squares	3.4× larger	4.1× larger
Docks that would move to match demand share, by station	27.2%	29.3%
Same, by district (25 districts; 95% CI)	9.9% (8.9 to 11.4)	9.4% (8.3 to 11.3)
Stations in the fit	2,472	2,470

WHAT A FIXED FLEET COULD DO

- The fleet covers 68% (weekday) and 76% (weekend) of estimated need, so the plan moves bikes rather than adding them: 4,874 weekday and 4,206 weekend bikes.
- Stations that gain bikes had a 22.2% mean stockout rate; stations that give bikes, 2.4%.
- **Equity check.** Net moves are unrelated to a district's demand per station (rank correlation 0.05 weekday, −0.06 weekend). The largest net givers are Geumcheon, Dongdaemun, Guro and Jungnang (11 to 20% of current bikes); the largest receivers are Jung-gu, Jongno and Gangbuk (+25 to +50%). Moves use midnight stock, which can inflate central districts, so read this as a flag for review, not a verdict.
- Ratings: 367 HIGH, 4,492 MEDIUM, 83 LOW. Direction agreement with the base plan: 97.9% (turnover cap 12), 99.5% (no cap), 97.8% (stockout correction), 87.2% (no dock bound), 72.7% (demand-proportional H_0).

LIMITS

- One month, no weather or season. Demand model validation and truck-rebalancing logs are not available.
- Move size and stockout rate correlate at 0.82, partly by construction, so this is not independent proof. Move size and predicted demand correlate at 0.025 (pooled rank correlation).
- 1,889 of 4,942 targets exceed nominal docks; the 1.5× dock limit is an assumption. Double ML was not run.

Updated 17 October 2026. Interactive site, system maps report, data and code: see the project package. Figures are recomputed from ttareungi_FINAL_recommendations_v2_202509.csv; the Wald statistics match the site and report values.